

NeuroCompute — The Operating System for Neuromorphic AI Hardware

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Category: AI Infrastructure / DeepTech

Market Size: \$180B (AI Hardware + Edge Computing)

Investment Stage: Seed to Series A

Executive Summary

The AI hardware revolution is hitting a wall. GPUs consume 85% of data center energy. Inference costs are exploding. The next frontier isn't bigger GPUs—it's brain-inspired neuromorphic computing that's 1000x more energy efficient.

NeuroCompute is the operating system that makes neuromorphic chips accessible to every developer. We're doing for neuromorphic computing what CUDA did for GPUs—creating the abstraction layer that unlocks a \$180B market transition.

The Problem

1. The GPU Energy Crisis

- Data centers now consume 4% of global electricity (up from 1% in 2020)
- A single GPT-6 query costs 100x more energy than a Google search
- AI infrastructure is hitting physical limits: cooling, power, sustainability
- Enterprise AI projects are being killed due to inference costs

2. Neuromorphic Hardware is Ready, But Unusable

- Intel's Loihi 3, IBM's NorthPole, Rain Neuromorphics chips shipping at scale
- 100-1000x more energy efficient than GPUs for inference
- **Problem:** Every chip has different programming models, no standard interfaces
- Result: Only PhDs can use this hardware; 99.9% of developers locked out

3. The Edge AI Imperative

- 75% of enterprise data will be processed at the edge by 2027
- Autonomous vehicles, robots, AR glasses need on-device AI
- Current edge AI is crippled by power constraints
- Neuromorphic is the only path to always-on, battery-powered AI

4. No Unified Stack Exists

- GPU world has CUDA, PyTorch, cloud providers
 - Neuromorphic world has fragmented SDKs, no cloud deployment, no MLOps
 - Companies building neuromorphic solutions spend 80% on infrastructure, 20% on product
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The Solution

NeuroCompute: The Universal Neuromorphic OS

NEUROCOMPUTE STACK

Developer Interface

NeuroScript: Unified language for spiking networks
Model Zoo: Pre-trained SNNs for common tasks
Conversion Tools: PyTorch → Neuromorphic (1 click)

Runtime & Orchestration

NeuroKernel: Unified runtime across all chips
Event Router: Spike-based communication layer
Resource Manager: Adaptive compute allocation

Hardware Abstraction Layer

Intel	IBM	Rain	BrainChip
Loihi 3	NorthPole	NeuronX	Akida 2.0

Core Capabilities

1. NeuroScript — Write Once, Run Anywhere

```
# NeuroScript: A unified language for neuromorphic computing
from neurocompute import SNN, SpikingLayer, deploy

# Define a spiking neural network
model = SNN([
    SpikingLayer(input=784, neurons=256, threshold=0.5),
    SpikingLayer(neurons=128, threshold=0.3),
    SpikingLayer(neurons=10, threshold=0.2)
])

# Train on standard PyTorch dataset
model.train(mnist, epochs=10)

# Deploy to ANY neuromorphic chip - NeuroCompute handles the rest
deploy(model, target="edge", chip="auto-select")
```

2. Model Conversion Engine

- **One-click conversion** from PyTorch/TensorFlow to spiking neural networks
- Automatic optimization for target hardware
- Maintains 95%+ accuracy during ANN-to-SNN conversion
- Supports transformers, CNNs, RNNs, and hybrid architectures

3. NeuroCloud — Neuromorphic Compute as a Service

- First cloud platform for neuromorphic inference
- Pay-per-spike pricing (1000x cheaper than GPU inference)
- Global edge deployment to 50+ regions

- Hybrid deployment: cloud neuromorphic + edge neuromorphic

4. Real-Time Event Processing

- Native support for event-driven, sparse data
 - Perfect for: sensors, cameras, audio, time-series
 - Microsecond latency for real-time applications
 - Continuous learning without catastrophic forgetting
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Why Now?

1. Hardware Inflection Point (2025-2026)

- Intel Loihi 3: 1M neurons per chip, available for commercial deployment
- IBM NorthPole: First production neuromorphic chip for data centers
- Rain Neuromorphics: Analog neuromorphic chips hitting \$10/chip price point
- BrainChip Akida 2.0: Edge neuromorphic in consumer devices

2. Energy Economics Forcing the Shift

- EU Carbon Border Adjustment taxing data center emissions
- California mandating data center energy efficiency by 2027
- Hyperscalers (AWS, Azure, Google) desperate for GPU alternatives
- Neuromorphic delivers same inference at 0.1% of energy cost

3. Edge AI Demand Explosion

- Apple, Samsung, Meta need always-on AI in glasses and wearables
- Autonomous vehicles need 1000x current inference at 10x less power
- Industrial IoT requires real-time AI without cloud connectivity
- Neuromorphic is the ONLY solution for battery-powered inference

4. Developer Ecosystem is Missing

- 10M developers know PyTorch
 - <5,000 can program neuromorphic chips
 - NeuroCompute bridges this 2000x gap
 - First-mover advantage in defining the standard
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Market Opportunity

Total Addressable Market: \$180B by 2030

Segment	2026	2030	CAGR
AI Hardware (chips)	\$45B	\$120B	28%
Edge AI Infrastructure	\$18B	\$40B	22%
AI Cloud Services	\$80B	\$180B	22%
Neuromorphic Share	\$2B	\$35B	104%

Serviceable Addressable Market: \$35B

- Enterprise neuromorphic deployment
- Edge AI at scale
- Neuromorphic cloud services

Serviceable Obtainable Market (Year 3): \$500M

- 5,000 enterprise customers $\times$ \$100K average contract
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Business Model

1. Platform Subscription (Primary)

Tier	Price	Features
Developer	Free	NeuroScript IDE, 1M spike-hours/month
Team	\$500/mo	Collaboration, 100M spike-hours, priority support
Enterprise	\$10K/mo	Unlimited, SLA, dedicated support, on-prem option

2. NeuroCloud Consumption

- **\$0.0001 per million spikes** (vs. \$0.10 per GPU inference)
- Reserved capacity discounts (40% savings)
- Commitment tiers for large deployments

3. Hardware Partnerships

- Revenue share with chip manufacturers
- 15% of chip sales enabled by NeuroCompute
- Embedded licensing for edge devices

4. Professional Services

- Model conversion services: \$50K-500K per engagement
- Custom deployment: \$200K-2M enterprise deals
- Training and certification programs

Revenue Projection

Year	ARR	Customers	Team
1	\$2M	50	15
2	\$15M	400	45
3	\$80M	2,500	120
4	\$250M	8,000	300
5	\$600M	20,000	500

Competitive Landscape

Direct Competitors

Player	Approach	Weakness
Intel LAVA	Loihi-only framework	Single-vendor lock-in
IBM SyNAPSE	Research-focused	No commercial deployment
BrainChip MetaTF	Akida-specific	Limited to one chip family
Nengo (Waterloo)	Academic simulator	Not production-ready

Indirect Competition

- **NVIDIA CUDA:** GPU-only, no neuromorphic support
- **Apache TVM:** Compiler focus, not neuromorphic-native
- **Edge Impulse:** Limited to tiny ML, not neuromorphic

NeuroCompute Advantages

1. **Multi-vendor:** Works with ALL neuromorphic chips
2. **Production-ready:** Enterprise SLAs, not academic software
3. **Developer-first:** 10-minute time-to-hello-world
4. **Cloud-native:** First neuromorphic cloud platform
5. **Ecosystem:** Model zoo, community, marketplace

Go-to-Market Strategy

Phase 1: Developer Adoption (Months 1-12)

- Launch free tier with NeuroScript and simulator
- Partner with Intel, IBM, Rain for hardware access
- Open-source core tools to build community
- Target: 50,000 developers, 500 active projects

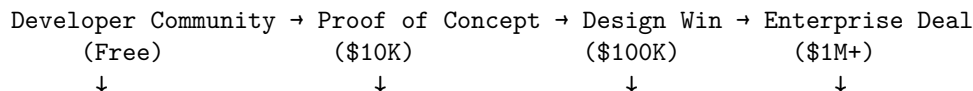
Phase 2: Enterprise Landing (Months 6-24)

- Focus on energy-constrained use cases:
 - Autonomous vehicle perception
 - Industrial IoT anomaly detection
 - AR/VR always-on AI
 - Satellite edge processing
- Land 50 design wins with Fortune 500

Phase 3: Platform Expansion (Months 18-36)

- Launch NeuroCloud in AWS, Azure, GCP
- Model marketplace for pre-trained SNNs
- Certification program for partners
- International expansion: EU, APAC

Customer Acquisition Strategy



50,000 1,000 200 50

Technology Architecture

Core IP

1. NeuroCompile — Unified Compilation Pipeline

PyTorch Model → Graph Optimization → Spike Encoding →
Hardware Mapping → Runtime Binary → Deployment

- Automatic neuron partitioning across chip cores
- Spike-timing optimization for accuracy
- Hardware-specific optimizations maintained in abstraction

2. NeuroKernel — Universal Runtime

- Lightweight runtime (<1MB) for all neuromorphic chips
- Event-driven scheduler for spike routing
- Memory-efficient state management
- Hot-swap model updates without restart

3. NeuroSync — Distributed Neuromorphic Computing

- First framework for multi-chip spiking networks
 - Scales to millions of neurons across chips
 - Synchronization protocols for distributed learning
 - Enables neuromorphic “clusters” like GPU clusters today
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Team Requirements

Founding Team (Raise \$5M Seed)

- **CEO:** Enterprise platform sales + AI background
- **CTO:** Neuromorphic computing PhD + systems engineering
- **VP Engineering:** CUDA/runtime systems experience
- **VP Product:** Developer tools product leader

Key Hires (Post-Seed)

- Compiler engineers (LLVM, MLIR experience)
- Hardware integration leads (one per chip vendor)
- Developer relations team
- Enterprise solutions architects

Advisory Board

- Neuromorphic pioneers from Intel, IBM, Stanford
 - AI infrastructure executives from hyperscalers
 - Enterprise CIOs from target verticals
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Financial Projections

5-Year P&L Summary

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	\$2M	\$15M	\$80M	\$250M	\$600M
Gross Margin	75%	78%	80%	82%	85%
Operating Expenses	\$8M	\$25M	\$70M	\$180M	\$350M
EBITDA	-\$6M	-\$13M	-\$6M	\$25M	\$160M
Headcount	15	45	120	300	500

Funding Roadmap

Round	Amount	Timing	Milestones
Seed	\$5M	Now	Product launch, 3 chip integrations
Series A	\$25M	Month 12	10K developers, \$2M ARR
Series B	\$80M	Month 24	50 enterprise customers, \$15M ARR
Series C	\$200M	Month 42	Market leadership, path to profitability

Risks and Mitigations

Risk	Probability	Impact	Mitigation
Slow neuromorphic adoption	Medium	High	Focus on energy-crisis verticals first
Chip vendor competition	Medium	Medium	Multi-vendor strategy, open standards
Technical complexity	High	Medium	Hire top compiler talent, phased rollout
GPU efficiency improvements	Low	High	Neuromorphic benefits grow with scale
Enterprise sales cycle	Medium	Medium	Land with innovation teams, expand

The Ask

Seed Round: \$5M

Use of Funds: - Engineering (60%): \$3M - Core platform development - 3 chip integrations (Intel, IBM, Rain) - Developer tools and documentation - Go-to-Market (25%): \$1.25M - Developer community building - Enterprise pilot program - Marketing and content - Operations (15%): \$0.75M - Team building - Infrastructure - Legal/IP

12-Month Milestones: - [] NeuroScript 1.0 launched - [] 10,000 developers on platform - [] 3 chip integrations complete - [] 10 enterprise pilots - [] \$500K in committed ARR

Why We Win

1. First-Mover in Defining the Standard

Just as NVIDIA's CUDA became the standard for GPU computing, NeuroCompute will become the standard for neuromorphic. We're building during the hardware transition window.

2. Platform Economics

Every neuromorphic chip sold needs NeuroCompute. As the ecosystem grows, we capture value from: - Developer tools and training - Cloud inference revenue - Hardware revenue share - Enterprise deployments

3. The Energy Imperative

The world MUST move beyond GPUs. Energy costs, carbon regulations, and physics demand neuromorphic computing. We're building the bridge.

4. Network Effects

- More developers → more models
 - More models → more enterprise adoption
 - More adoption → more chip partnerships
 - More chips → more developers
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Vision: The Neuromorphic Era

By 2030, neuromorphic computing will process 50% of edge AI workloads. Data centers will run hybrid GPU-neuromorphic architectures. Every autonomous system—from robots to vehicles to buildings—will think in spikes.

NeuroCompute will be the operating system that makes this possible.

We're not building a company. We're building the infrastructure for brain-inspired computing.

Contact

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"The future of AI isn't bigger GPUs. It's smarter silicon that thinks like the brain."